

$$A = [2; -7]$$

$$B = [-3; 1]$$

$$x = 2 - 5t$$

$$y = -7 + 8t$$

$$\begin{aligned} x &= a_1 + u_1 \cdot t \\ y &= a_2 + u_2 \cdot t \end{aligned}$$

$$\overrightarrow{AB} = (-5; 8)$$

$$A = [1; -3]$$

$$B = [-2; 4]$$

$$x = 1 - 3t$$

$$y = -3 + 7t$$

$$\overrightarrow{AB} = (-3; 7)$$

$$C = [c_1; 1]$$

$$D = [-5; d_2]$$

$$E = [e_1; -2]$$

$$x = 1 - 3t$$

$$y = -3 + 7t$$

$$C \begin{cases} c_1 = 1 - 3t \Rightarrow 1 - 3 \cdot \frac{4}{7} = 1 - \frac{12}{7} = \frac{7}{7} - \frac{12}{7} = \underline{\underline{\frac{-5}{7}}} \\ 1 = -3 + 7t \Rightarrow t = \frac{4}{7} \end{cases}$$

$$D \begin{cases} -5 = 1 - 3t \Rightarrow t = 2 \\ d_2 = -3 + 7t \Rightarrow -3 + 14 = \underline{\underline{11}} \end{cases}$$

$$E \begin{cases} c_1 = 1 - 3t \Rightarrow \frac{7}{7} - \frac{3}{7} = \underline{\underline{\frac{4}{7}}} \\ -2 = -3 + 7t \Rightarrow t = \frac{1}{7} \end{cases}$$

$$A = [-1; 2]$$

$$\vec{u} = (3; 4) \perp p$$

$$ax + by + c = 0$$

$$3x + 4y + c = 0$$

$$A \in p: 3 \cdot (-1) + 4 \cdot 2 + c = 0$$

$$\underline{\underline{c = -5}}$$

$$ax + by + c = 0$$

$$p: 3x + 4y - 5 = 0$$

$$B = [-3; 2]$$

$$\vec{v} = (2; 1) \perp p$$

$$ax + by + c = 0$$

$$2x + 1y + c = 0$$

$$B \in p: 2 \cdot (-3) + 2 + c = 0$$

$$c = 4$$

$$p: 2x + 1y + 4 = 0$$

$$\begin{array}{l} \vec{h} \parallel (a; b) \\ \vec{h} \perp (-b; a) \end{array}$$

$$A = [3; -1]$$

$$\vec{u} = (3; -2) \parallel p$$

$$ax + by + c = 0$$

$$2x + 3y + c = 0$$

$$A \in p: 2 \cdot 3 + 3 \cdot (-1) + c = 0$$

$$c = -3$$

$$p: 2x + 3y - 3 = 0$$

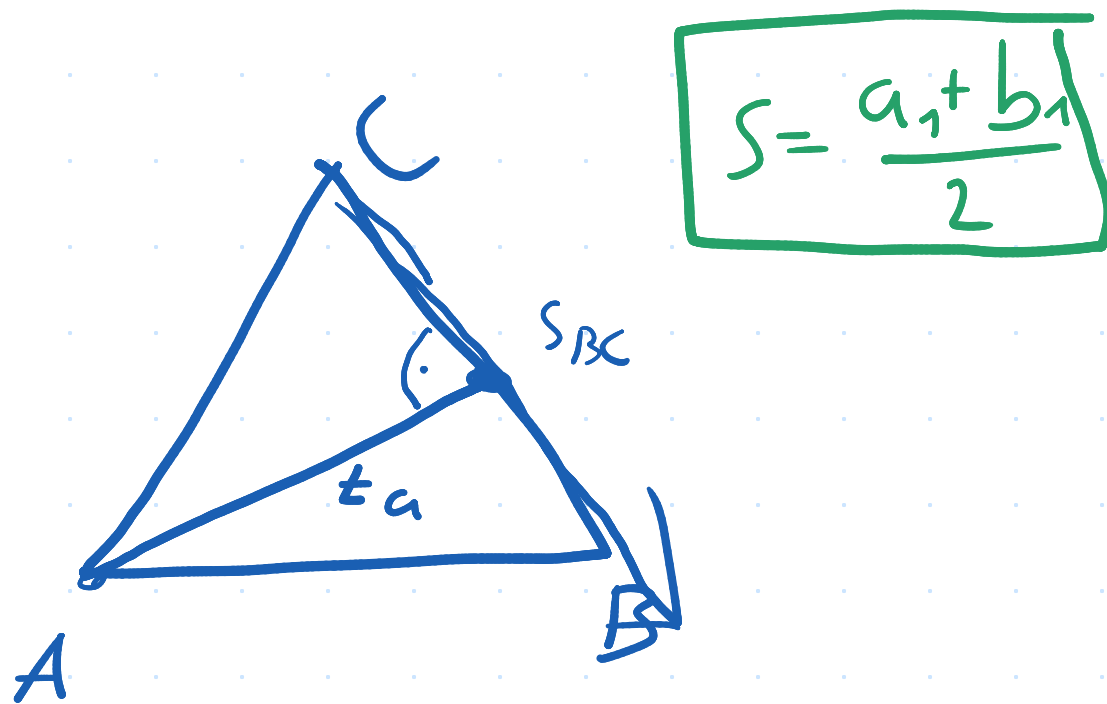
$$\vec{u} = (2; 3) \perp p$$

$\triangle ABC$

$$A = [-4; 2]$$

$$B = [4; 0]$$

$$C = [2; 6]$$



para. vce. t_a

$$S_{BC} = [3; 3]$$

$$\overrightarrow{AS_{BC}} = (7; 1)$$

$$x = -4 + 7t$$

$$y = 2 + t$$

obec. vce. t_a

$$\overrightarrow{AS} = (7; 1) \perp (-1; 7) = \vec{n}$$

$$ax + by + c = 0$$

$$-x + 7y + c = 0$$

$$A \in P: 4 + 14 + c = 0$$

$$c = -18$$

$\triangle ABC$

$$A = [6; 2]$$

$$B = [-2; 4]$$

$$C = [-2; 0]$$

obec. vce. t_a

$$S_{BC} = [-2; 2]$$

$$\overrightarrow{AS_{BC}} = (-8; 0)$$

$$\vec{n} = (0; -8)$$

$$ax + by + c = 0$$

$$-8y + c = 0$$

$$A \in P: -16 + c = 0$$

$$c = 16$$

$$P: -8y + 16 = 0$$